Linear Algebra and Its Applications

Section 1.4: The Matrix Equation Ax = b

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

Contents

1.4 The matrix equation $Ax = b$ 3

Practice problem 1.4.P1: Read coefficients from a known solution 3

Practice problem 1.4.P2: Check additivity by both calculations 3

Exercise 1.4.1: Check dimensions before multiplying 3

Exercise 1.4.2: A column matrix is not a row matrix 3

Exercise 1.4.3: Compute one product in two ways 4

Exercise 1.4.4: Two rows give two output entries 4

Exercise 1.4.5: Expand the matrix equation into columns 4

Exercise 1.4.6: A four-entry output from two columns 4

Exercise 1.4.7: Assemble the coefficient vectors as columns 5

Exercise 1.4.8: Four scalar weights and a two-dimensional target 5

Exercise 1.4.9: Keep the missing coefficient equal to zero 5

Exercise 1.4.10: Three equations become three coordinates 5
Exercise 1.4.11: Solve an augmented system and check its vector 6
Exercise 1.4.12: A second exact augmented reduction 6
Exercise 1.4.13: Membership in a plane means consistency 6
Exercise 1.4.14: A contradictory coordinate rules out membership 6
Exercise 1.4.15: Describe every attainable right-hand side 7
Exercise 1.4.16: A condition on three coordinates of the right side 7
Exercise 1.4.17: Count pivot rows of the specified matrix 7
Exercise 1.4.18: Distinguish the target space from the number of columns 7
Exercise 1.4.19: Translate the pivot calculation into a span statement 8
Exercise 1.4.20: The span and solvability questions are equivalent 8
Exercise 1.4.21: Three columns cannot fill four target directions 8
Exercise 1.4.22: Three suitable vectors do span three-dimensional space 8
Exercise 1.4.23: Six true-or-false statements about the matrix equation 8
Exercise 1.4.24: Six further equivalences and counterexamples 9
Exercise 1.4.25: Read a supplied product without repeating elimination 9
Exercise 1.4.26: Rearrange a known linear relation 9
Exercise 1.4.27: Name every vector and every coefficient 9
Exercise 1.4.28: Package a symbolic vector equation 10
Exercise 1.4.29: Construct a spanning matrix outside echelon form 10
Exercise 1.4.30: Construct a nonspanning matrix outside echelon form 10
Exercise 1.4.31: More rows than columns leave a missing pivot row 10
Exercise 1.4.32: Too few spanning vectors 10
Exercise 1.4.33: A unique solution fixes the reduced form 11
Exercise 1.4.34: Add solutions to reach the sum of their images 11
Exercise 1.4.35: Scale a known input 11
Exercise 1.4.36: A unique square-system solution forces spanning 11
Exercise 1.4.37: Use exact row reduction to test four-dimensional spanning 11
Exercise 1.4.38: Use exact row reduction to test four-dimensional spanning 12
Exercise 1.4.39: Use exact row reduction to test four-dimensional spanning 12
Exercise 1.4.40: Use exact row reduction to test four-dimensional spanning 12
Exercise 1.4.41: Delete a redundant column while retaining the span 13
Exercise 1.4.42: One deletion is possible but two are not 13

1.4 The matrix equation $Ax = b$

A matrix–vector product is a weighted sum of the matrix's columns. If $A = [a_1 \ \cdots \ a_n]$ and $x = (x_1, \ldots, x_n)^T$, then

$$Ax = x_1a_1 + \cdots + x_na_n.$$

Thus Ax is defined only when x has one entry for each column of A. The row rule gives the same product: entry i is $a_{i1}x_1 + \cdots + a_{in}x_n$. A vector b belongs to the span of the columns exactly when $Ax = b$ is consistent. Theorem 4 says the columns span the whole target $\mathbb{R}^m$ exactly when A has a pivot in every row. A pivot in an augmented column does not count as a pivot in A.

Practice problem 1.4.P1: Read coefficients from a known solution

The columns are $a_1 = (1, -3, 4)^T$, $a_2 = (5, 1, -8)^T$, $a_3 = (-2, 9, -1)^T$, and $a_4 = (0, -5, 7)^T$. The given solution has entries $(3, -2, 0, -4)^T$, so the definition gives

$$3a_1 - 2a_2 + 0a_3 - 4a_4 = (3, -9, 12)^T + (-10, -2, 16)^T + (0, 20, -28)^T = (-7, 9, 0)^T = b.$$

The coefficients are the entries of the solution, not row entries of A.

Practice problem 1.4.P2: Check additivity by both calculations

Here $A = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$, $u = (4, -1)^T$, and $v = (-3, 5)^T$. First $u + v = (1, 4)^T$, so

$$A(u + v) = (2 + 20, 3 + 4)^T = (22, 7)^T.$$

Separately $Au = (8 - 5, 12 - 1)^T = (3, 11)^T$ and $Av = (-6 + 25, -9 + 5)^T = (19, -4)^T$. Their sum is again $(22, 7)^T$. This checks $A(u + v) = Au + Av$ with the actual data.

Exercise 1.4.1: Check dimensions before multiplying

What we must show. Compute the indicated product, if possible, in both requested ways. The matrix $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix}$ has two columns, but $(3, -2, 7)^T$ has three entries. In the column definition there are only two columns for three proposed weights. In the row rule a row has two entries and cannot be paired with a three-entry vector.

Conclusion. The product is undefined; no numerical result can be assigned.

Exercise 1.4.2: A column matrix is not a row matrix

The matrix $(1, 3, 1)^T$ has size 3×1, whereas $(5, -1)^T$ has size 2×1. The inner sizes 1 and 2 differ. The column definition would need one scalar, and each row contains only one entry.

Conclusion. The product is undefined by either rule.

Exercise 1.4.3: Compute one product in two ways

(a) Weight the two columns.

$$-2\begin{bmatrix}1\\-3\\1\end{bmatrix}+3\begin{bmatrix}2\\1\\6\end{bmatrix}=\begin{bmatrix}-2+6\\6+3\\-2+18\end{bmatrix}=\begin{bmatrix}4\\9\\16\end{bmatrix}.$$

(b) Use the rows.

The three dot products are $1(-2)+2(3)=4$, $(-3)(-2)+1(3)=9$, and $1(-2)+6(3)=16$.

Conclusion. Both rules give $(4,9,16)^T$.

Exercise 1.4.4: Two rows give two output entries

(a) Use the column definition.

$$1\begin{bmatrix}1\\3\end{bmatrix}+2\begin{bmatrix}3\\2\end{bmatrix}+1\begin{bmatrix}-4\\1\end{bmatrix}=\begin{bmatrix}1+6-4\\3+4+1\end{bmatrix}=\begin{bmatrix}3\\8\end{bmatrix}.$$

(b) Use the row rule.

The first entry is $1(1)+3(2)-4(1)=3$; the second is $3(1)+2(2)+1(1)=8$. Both entries have now been checked independently.

Exercise 1.4.5: Expand the matrix equation into columns

Read each column vertically and use the corresponding entry of the multiplier vector as its weight:

$$2\begin{bmatrix}1\\-2\end{bmatrix}-\begin{bmatrix}2\\-3\end{bmatrix}+\begin{bmatrix}-3\\1\end{bmatrix}-\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}-4\\1\end{bmatrix}.$$

For a direct check, the first coordinate is $2-2-3-1=-4$ and the second is $-4+3+1+1=1$.

Exercise 1.4.6: A four-entry output from two columns

The equivalent vector equation is

$$-3\begin{bmatrix}2\\3\\8\\-2\end{bmatrix}+5\begin{bmatrix}-3\\2\\-5\\1\end{bmatrix}=\begin{bmatrix}-21\\1\\-49\\11\end{bmatrix}.$$

The coordinates are $-6-15=-21$, $-9+10=1$, $-24-25=-49$, and $6+5=11$.

Exercise 1.4.7: Assemble the coefficient vectors as columns

Place the coefficient of x_1 in column 1, that of x_2 in column 2, and that of x_3 in column 3. The result is

$$\begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -5 & 0 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}.$$

The output has four entries because each original vector lies in $\mathbb{R}^4$.

Exercise 1.4.8: Four scalar weights and a two-dimensional target

The same column convention gives

$$\begin{bmatrix} 2 & -1 & -4 & 0 \\ -4 & 5 & 3 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}.$$

There are four columns, one for each weight, and two rows, one for each coordinate equation.

Exercise 1.4.9: Keep the missing coefficient equal to zero

The coefficient of x_1 in the second equation is zero. Therefore

$$x_1 \begin{bmatrix} 5 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix},$$

and the matrix equation is

$$\begin{bmatrix} 5 & 1 & -3 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 0 \end{bmatrix}.$$

Exercise 1.4.10: Three equations become three coordinates

The vector equation is

$$x_1 \begin{bmatrix} 4 \\ 5 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

Equivalently,

$$\begin{bmatrix} 4 & -1 \\ 5 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \\ 1 \end{bmatrix}.$$

No equation has been discarded in changing notation.

Exercise 1.4.11: Solve an augmented system and check its vector

Form and reduce the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 1 & 5 & 2 & 4 \\ -3 & -7 & 6 & 12 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3+3R_1]{R_2 \leftarrow R_2-R_1} \left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 0 & 2 & 6 & 6 \\ 0 & 2 & -6 & 6 \end{array}\right].$$

Replace R_3 by $R_3 - R_2$ to obtain $-12x_3 = 0$. Thus $x_3 = 0$. The second row gives $2x_2 + 6x_3 = 6$, hence $x_2 = 3$. The first gives $x_1 + 9 = -2$, hence $x_1 = -11$.

Substitute into the original rows.

They give $-11 + 9 = -2$, $-11 + 15 = 4$, and $33 - 21 = 12$.

Conclusion. The solution vector is $x = (-11, 3, 0)^T$.

Exercise 1.4.12: A second exact augmented reduction

Start with

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ -3 & -4 & 2 & 2 \\ 5 & 2 & 3 & -3 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3-5R_1]{R_2 \leftarrow R_2+3R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 2 & -1 & 5 \\ 0 & -8 & 8 & -8 \end{array}\right].$$

Now $R_3 \leftarrow R_3 + 4R_2$ gives $4x_3 = 12$, so $x_3 = 3$. Then $2x_2 - 3 = 5$ gives $x_2 = 4$, and $x_1 + 8 - 3 = 1$ gives $x_1 = -4$. The original products are $-4 + 8 - 3 = 1$, $12 - 16 + 6 = 2$, and $-20 + 8 + 9 = -3$.

Conclusion. $x = (-4, 4, 3)^T$.

Exercise 1.4.13: Membership in a plane means consistency

What we must show. Decide whether $u = (0, 4, 4)^T$ is a linear combination of $(3, -2, 1)^T$ and $(-5, 6, 1)^T$. The equations for the two weights s, t are $3s - 5t = 0$, $-2s + 6t = 4$, and $s + t = 4$. The last equation gives $s = 4 - t$. Substitution into the first gives $12 - 8t = 0$, so $t = 3/2$ and $s = 5/2$. The second equation checks: $-5 + 9 = 4$. Hence

$$u = \tfrac{5}{2}(3, -2, 1)^T + \tfrac{3}{2}(-5, 6, 1)^T.$$

Conclusion. Yes: the explicit weights show that u lies in the plane.

Exercise 1.4.14: A contradictory coordinate rules out membership

The proposed equation $Ax = u$ gives

$$2x_1 + 5x_2 - x_3 = 4, \qquad x_2 - x_3 = -1, \qquad x_1 + 2x_2 = 4.$$

From the last two equations $x_3 = x_2 + 1$ and $x_1 = 4 - 2x_2$. The left side of the first would then be $2(4 - 2x_2) + 5x_2 - (x_2 + 1) = 7$, not 4.

Conclusion. No. The system is inconsistent, so u is outside the column span.

Exercise 1.4.15: Describe every attainable right-hand side

For $A = \begin{bmatrix} 3 & -1 \\ -9 & 3 \end{bmatrix}$, the second row is -3 times the first. Eliminating it in $[A \mid b]$ gives the equation $0 = b_2 + 3b_1$. Thus consistency is equivalent to $b_2 = -3b_1$. This is also sufficient: the first column $(3, -9)^T$ alone reaches $(b_1, -3b_1)^T$ with weight $b_1/3$.

Conclusion. The attainable vectors are exactly $t(1, -3)^T$, $t \in \mathbb{R}$; for example $(0, 1)^T$ is not attainable, so not every b works.

Exercise 1.4.16: A condition on three coordinates of the right side

For the given matrix, reduce symbolically:

$$\left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ -2 & 2 & 0 & b_2 \\ 4 & -1 & 3 & b_3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & -2 & -1 & b_1 \\ 0 & -2 & -2 & b_2 + 2b_1 \\ 0 & 7 & 7 & b_3 - 4b_1 \end{array}\right].$$

These steps use $R_2 + 2R_1$ and $R_3 - 4R_1$. Replacing the last row by $2R_3 + 7R_2$ gives $0 = 6b_1 + 7b_2 + 2b_3$. If this condition holds, the remaining two equations have pivots and can be solved, with one free variable. If it fails, the last row is a contradiction.

Conclusion. The attainable right sides form the plane $\{b : 6b_1 + 7b_2 + 2b_3 = 0\}$, not all of $\mathbb{R}^3$.

Exercise 1.4.17: Count pivot rows of the specified matrix

For the matrix A used in Exercises 17 and 19, exact elimination gives

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are three pivot rows, in columns 1, 2, and 4. The fourth row of the coefficient matrix has no pivot. By Theorem 4, $Ax = b$ is solvable for every $b \in \mathbb{R}^4$ if and only if A has a pivot in every row.

Conclusion. Three rows contain pivots; some right sides have no solution.

Exercise 1.4.18: Distinguish the target space from the number of columns

For the matrix B used here and in Exercise 20,

$$B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -11 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There are only three pivots and one zero row. Theorem 4 therefore shows that its columns do not span $\mathbb{R}^4$. They do not span $\mathbb{R}^3$ either: the columns themselves have four entries and belong to $\mathbb{R}^4$, so $\mathbb{R}^3$ is not their ambient space.

Conclusion. Not every vector of $\mathbb{R}^4$ is a combination of the columns; the proposed $\mathbb{R}^3$ claim also fails.

Exercise 1.4.19: Translate the pivot calculation into a span statement

The reduction displayed in Exercise 1.4.17 has a zero fourth row and only three pivots. Theorem 4 says this is equivalent to failure of the columns to span $\mathbb{R}^4$. A right side whose transformed fourth entry is nonzero gives an inconsistent augmented system.

Conclusion. No to both questions: not every vector is a column combination, and the column span is a proper subset of $\mathbb{R}^4$.

Exercise 1.4.20: The span and solvability questions are equivalent

Exercise 1.4.18 exhibits three pivot rows for B, not four. By Theorem 4, the following are equivalent: the columns span $\mathbb{R}^4$; $Bx = y$ is consistent for every $y \in \mathbb{R}^4$; and B has a pivot in every row. The last condition fails.

Conclusion. The columns do not span $\mathbb{R}^4$, and some y give no solution.

Exercise 1.4.21: Three columns cannot fill four target directions

A 4×3 matrix formed from these three vectors has at most three pivot positions, because each pivot requires a different column. It therefore cannot have a pivot in every one of its four rows. Theorem 4 excludes spanning $\mathbb{R}^4$. More concretely each vector's coordinate sum is zero; every linear combination also has coordinate sum zero, whereas $(1, 0, 0, 0)^T$ does not.

Conclusion. The three vectors do not span $\mathbb{R}^4$.

Exercise 1.4.22: Three suitable vectors do span three-dimensional space

Put the vectors into columns:

$$V = \begin{bmatrix} 0 & 0 & 4 \\ 0 & -3 & -2 \\ -3 & 9 & -6 \end{bmatrix}.$$

Swap rows 1 and 3. The resulting matrix has pivots $-3, -3, 4$ in successively later columns, so it has three pivot rows. Theorem 4 implies its columns span $\mathbb{R}^3$. One can also solve $Vc = b$ directly: $c_3 = b_1/4$, then $c_2 = -(b_2 + 2c_3)/3$, and finally $c_1 = (9c_2 - 6c_3 - b_3)/3$. These formulas work for every b.

Conclusion. Yes, the vectors span $\mathbb{R}^3$.

Exercise 1.4.23: Six true-or-false statements about the matrix equation

(a) **False.** $Ax = b$ is called a matrix equation. Its equivalent column-sum expression is a vector equation.

(b) **True.** By definition, Ax runs through precisely the linear combinations of the columns; existence of a suitable x is the claim.

(c) **False.** A pivot in every row of $[A \mid b]$ can include a pivot in the last column. For instance $[0 \mid 1]$ has a pivot in its only row but represents $0x = 1$, an inconsistent equation.

(d) **True.** The first entry is $a_{11}x_1 + \cdots + a_{1n}x_n$, a sum of scalar products.

(e) **True.** This is the column-span equivalence in Theorem 4.

(f) **True.** If A had a pivot in every row, Theorem 4 would make every right side attainable. The stated inconsistent right side contradicts that possibility.

Exercise 1.4.24: Six further equivalences and counterexamples

(a) **True.** Expanding Ax into columns changes notation, not the equations or their solutions.

(b) **True.** A solution gives $b = x_1a_1 + \cdots + x_na_n$, which is exactly membership in the column span.

(c) **True.** Place the vectors being combined into the columns of A and their weights into x.

(d) **False.** A pivot in every coefficient row guarantees *consistency* for every right side, not inconsistency.

(e) **True.** Both descriptions use the coefficient matrix $A = [a_1\ a_2\ a_3]$ and the same right side b.

(f) **False.** Failure to span means that *some* b lies outside the span and gives an inconsistent equation. The right side 0 is always consistent, since $x = 0$ solves $Ax = 0$.

Exercise 1.4.25: Read a supplied product without repeating elimination

The given multiplier vector is $(-3, -1, 2)^T$. The definition of matrix multiplication therefore already states

$$(-7, -3, 10)^T = -3(4, 5, -6)^T - (-3, -2, 2)^T + 2(1, 5, -3)^T.$$

Conclusion. $c_1 = -3$, $c_2 = -1$, and $c_3 = 2$.

Exercise 1.4.26: Rearrange a known linear relation

The supplied equality $2u - 3v - w = 0$ rearranges to $w = 2u - 3v$. The displayed matrix has columns u and v, so its product with $(2, -3)^T$ is w. Checking the entries gives $14 - 9 = 5$, $4 - 3 = 1$, and $10 - 9 = 1$.

Conclusion. $x_1 = 2$ and $x_2 = -3$; no row operations are needed.

Exercise 1.4.27: Name every vector and every coefficient

Define

$$v_1 = (-3, 5)^T, \quad v_2 = (5, 8)^T, \quad v_3 = (-4, 1)^T, \quad v_4 = (9, -2)^T, \quad v_5 = (7, -4)^T.$$

Let $(c_1, c_2, c_3, c_4, c_5) = (-3, 1, 2, -1, 2)$ and $b = (11, -11)^T$. The symbolic vector equation is

$$c_1v_1 + c_2v_2 + c_3v_3 + c_4v_4 + c_5v_5 = b.$$

Its coordinate sums are $9 + 5 - 8 - 9 + 14 = 11$ and $-15 + 8 + 2 + 2 - 8 = -11$, checking the interpretation.

Exercise 1.4.28: Package a symbolic vector equation

Set $Q = [q_1\ q_2\ q_3]$, a 5×3 matrix, and $x = (x_1, x_2, x_3)^T$. The required matrix equation is $Qx = v$. Its five coordinates reproduce those of $x_1q_1 + x_2q_2 + x_3q_3 = v$, because a product is a weighted column sum.

Exercise 1.4.29: Construct a spanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The leading entry in row 2 lies left of that in row 1, so A is not in echelon form. Swapping rows 1 and 2 produces I_3, which has a pivot in every row. Theorem 4 therefore proves its columns span $\mathbb{R}^3$. Explicitly, $(b_1, b_2, b_3)^T$ is obtained with weights $(b_2, b_1, b_3)^T$.

Exercise 1.4.30: Construct a nonspanning matrix outside echelon form

Take

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The first two leading entries move left, so it is not echelon. Every product is $(x_2, x_1, x_2)^T$; its first and third entries agree. The vector $(1, 0, 0)^T$ violates that condition and cannot be obtained.

Conclusion. This matrix meets both requirements.

Exercise 1.4.31: More rows than columns leave a missing pivot row

A 3×2 matrix has at most two pivots, since there are only two columns. It cannot have a pivot in each of three rows. Theorem 4 then gives some $b \in \mathbb{R}^3$ for which $Ax = b$ is inconsistent. More generally, an $m \times n$ matrix with $m > n$ has at most $n < m$ pivots and hence cannot reach every vector of $\mathbb{R}^m$.

Conclusion. At least one target direction is missing whenever $m > n$.

Exercise 1.4.32: Too few spanning vectors

Place the three proposed vectors in the columns of a 4×3 matrix. At most three of its four rows can contain pivots, so Theorem 4 rules out spanning $\mathbb{R}^4$. In general n vectors in $\mathbb{R}^m$ with $n < m$ give an $m \times n$ matrix with too few columns to pivot in every row.

Conclusion. Neither three vectors in $\mathbb{R}^4$ nor $n < m$ vectors in $\mathbb{R}^m$ can span the whole target space.

Exercise 1.4.33: A unique solution fixes the reduced form

Since the given equation is consistent and has only one solution, none of its three unknowns can be free. Each of the three coefficient columns therefore contains a pivot. In reduced echelon form the pivot columns are unit vectors, with pivots successively lower. Thus the reduced echelon form of A is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

The fourth coefficient row must be zero because only three pivots are possible. Consistency ensures the corresponding transformed right-side entry is zero as well.

Exercise 1.4.34: Add solutions to reach the sum of their images

The additivity identity of Theorem 5 gives

$$A(u_1 + u_2) = Au_1 + Au_2 = v_1 + v_2 = w.$$

The vector $u_1 + u_2$ belongs to $\mathbb{R}^4$, the correct input space. It is an explicit solution to $Ax = w$, so the system is consistent.

Exercise 1.4.35: Scale a known input

The scalar-multiplication identity of Theorem 5 gives $A(5y) = 5Ay = 5z$. Thus $x = 5y$ is a solution to the new system. The factor multiplies the entire input vector, not just one entry.

Exercise 1.4.36: A unique square-system solution forces spanning

The unique solution means no unknown is free. All four coefficient columns have pivots, so the 4×4 matrix has four pivot rows. By Theorem 4 its columns span $\mathbb{R}^4$. Consistency for one special b alone would not be enough; uniqueness supplies the crucial four column pivots.

Exercise 1.4.37: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -\frac{93}{50} \\ 0 & 1 & 0 & \frac{53}{50} \\ 0 & 0 & 1 & -\frac{189}{50} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 3; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

Conclusion. The columns do not span $\mathbb{R}^4$.

Exercise 1.4.38: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 4 & -5 & -1 & 8 \\ 3 & -7 & -4 & 2 \\ 5 & -6 & -1 & 4 \\ 9 & 1 & 10 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are 1, 2, 4; there are 3 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. The last row has no pivot, so some right sides are unattainable.

Conclusion. The columns do not span $\mathbb{R}^4$.

Exercise 1.4.39: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 10 & -7 & 1 & 4 & 6 \\ -8 & 4 & -6 & -10 & -3 \\ -7 & 11 & -5 & -1 & -8 \\ 3 & -1 & 10 & 12 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 3, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

Conclusion. The columns do span $\mathbb{R}^4$.

Exercise 1.4.40: Use exact row reduction to test four-dimensional spanning

What we must show. Decide whether the columns of the given matrix span $\mathbb{R}^4$. These exercises explicitly allow matrix computation. Ordinary Gauss–Jordan elimination (swap a nonzero entry into each pivot position, scale the pivot to 1, then clear its column) gives the exact result

$$A = \begin{bmatrix} 5 & 11 & -6 & -7 & 12 \\ -7 & -3 & -4 & 6 & -9 \\ 11 & 5 & 6 & -9 & -3 \\ -3 & 4 & -7 & 2 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivot columns are 1, 2, 4, 5; there are 4 pivot rows. Theorem 4 requires a pivot in every one of the four rows, not merely in every nonzero row. That condition holds, so every right side in $\mathbb{R}^4$ is attainable.

Conclusion. The columns do span $\mathbb{R}^4$.

Exercise 1.4.41: Delete a redundant column while retaining the span

The pivot columns in the reduction of Exercise 1.4.39 are 1, 2, 3, and 5. These four original columns remain spanning after column 4 is deleted. The reduced form also displays the relation $a_4 = a_1+a_2+a_3$; row operations preserve a relation among columns because they multiply every column by the same invertible operation. Thus the deleted column adds no new span.

Conclusion. Delete column 4. The remaining four pivot columns span $\mathbb{R}^4$.

Exercise 1.4.42: One deletion is possible but two are not

Exercise 1.4.40 has pivot columns 1, 2, 4, and 5. Delete column 3; the remaining four original pivot columns still span $\mathbb{R}^4$. Indeed the reduced matrix shows $a_3 = a_1 - a_2$, so this column was redundant. Deleting two columns from the original five would leave only three vectors. Exercise 1.4.32 proves that three vectors cannot span $\mathbb{R}^4$.

Conclusion. Column 3 can be deleted; no pair of columns can be deleted while preserving the span.